

```

1 F:\Python\Training_Voice\.venv\Scripts\python.exe F:\Python\
  Training_Voice\training_voice.py
2 ☐ ☐ Using device: cuda
3 ☐ Loading dataset...
4 ☐ Dataset shape: (51077, 367)
5 ☐ Feature columns: 367
6 ☐ Classes: ['angry' 'boredom' 'disgust' 'fearful' 'happy' 'neutral' 'sad' '
  surprised']
7 ☐ Class Distribution: Counter({'neutral': 9662, 'angry': 9642, 'happy':
  9586, 'sad': 9577, 'surprised': 7384, 'fearful': 2584, 'disgust': 2561, '
  boredom': 81})
8 ☐ Filtering out 81 samples with rare classes
9 ☐ Splitting data...
10 ☐ ☐ Normalizing features...
11 ☐ Training set: 36844 samples
12 ☐ Validation set: 6502 samples
13 ☐ Test set: 7650 samples
14 ☐ Creating datasets with augmentation...
15 ☐ ☐ Initializing model...
16 ☐ Using Focal Loss for class imbalance
17
18 ☐ Starting training for 200 epochs...
19 =====
  =====
20 ☐ Epoch 001/200 (0.5%) | Train: 53.04% | Val: 64.46% | F1: 0.652 |
  Time: 3s | ETA: 12m 39s ☐ NEW BEST!
21 ☐ Epoch 002/200 (1.0%) | Train: 64.65% | Val: 67.99% | F1: 0.689 |
  Time: 3s | ETA: 11m 48s ☐ NEW BEST!
22 ☐ Epoch 003/200 (1.5%) | Train: 67.00% | Val: 69.69% | F1: 0.708 |
  Time: 3s | ETA: 11m 32s ☐ NEW BEST!
23 ☐ Epoch 004/200 (2.0%) | Train: 68.68% | Val: 71.21% | F1: 0.722 |
  Time: 3s | ETA: 11m 31s ☐ NEW BEST!
24 ☐ Epoch 005/200 (2.5%) | Train: 69.11% | Val: 73.35% | F1: 0.744 |
  Time: 3s | ETA: 11m 24s ☐ NEW BEST!
25 ☐ Epoch 007/200 (3.5%) | Train: 72.34% | Val: 74.92% | F1: 0.759 |
  Time: 3s | ETA: 11m 14s ☐ NEW BEST!
26 ☐ Epoch 008/200 (4.0%) | Train: 73.30% | Val: 75.21% | F1: 0.762 |
  Time: 3s | ETA: 11m 9s ☐ NEW BEST!
27 ☐ Epoch 009/200 (4.5%) | Train: 74.31% | Val: 76.47% | F1: 0.775 |
  Time: 3s | ETA: 11m 7s ☐ NEW BEST!
28 ☐ Epoch 010/200 (5.0%) | Train: 74.88% | Val: 76.13% | F1: 0.774 |
  Time: 3s | ETA: 11m 3s
29 ☐ Epoch 011/200 (5.5%) | Train: 75.96% | Val: 77.71% | F1: 0.787 |
  Time: 3s | ETA: 11m 0s ☐ NEW BEST!
30 ☐ Epoch 012/200 (6.0%) | Train: 76.14% | Val: 78.02% | F1: 0.789 |
  Time: 3s | ETA: 11m 3s ☐ NEW BEST!
31 ☐ Epoch 014/200 (7.0%) | Train: 78.02% | Val: 79.58% | F1: 0.807 |
  Time: 3s | ETA: 11m 3s ☐ NEW BEST!
32 ☐ Epoch 015/200 (7.5%) | Train: 78.85% | Val: 79.76% | F1: 0.807 |
  Time: 3s | ETA: 11m 3s ☐ NEW BEST!
33 ☐ Epoch 017/200 (8.5%) | Train: 79.97% | Val: 80.67% | F1: 0.817 |
  Time: 3s | ETA: 11m 1s ☐ NEW BEST!
34 ☐ Epoch 018/200 (9.0%) | Train: 79.93% | Val: 81.30% | F1: 0.822 |
  Time: 3s | ETA: 11m 0s ☐ NEW BEST!
35 ☐ Epoch 020/200 (10.0%) | Train: 81.45% | Val: 81.91% | F1: 0.828 |
  Time: 3s | ETA: 10m 49s ☐ NEW BEST!
36 ☐ Epoch 021/200 (10.5%) | Train: 81.64% | Val: 82.21% | F1: 0.833 |
  Time: 3s | ETA: 10m 45s ☐ NEW BEST!

```

37 ☐ Epoch 023/200 (11.5%) | Train: 82.46% | Val: 82.94% | F1: 0.838 |
 Time: 3s | ETA: 10m 35s ☐ NEW BEST!
 38 ☐ Epoch 026/200 (13.0%) | Train: 83.68% | Val: 83.21% | F1: 0.841 |
 Time: 3s | ETA: 10m 22s ☐ NEW BEST!
 39 ☐ Epoch 027/200 (13.5%) | Train: 83.93% | Val: 83.77% | F1: 0.845 |
 Time: 3s | ETA: 10m 18s ☐ NEW BEST!
 40 ☐ Epoch 028/200 (14.0%) | Train: 84.50% | Val: 83.87% | F1: 0.847 |
 Time: 3s | ETA: 10m 13s ☐ NEW BEST!
 41 ☐ Epoch 030/200 (15.0%) | Train: 85.03% | Val: 84.11% | F1: 0.847 |
 Time: 3s | ETA: 10m 5s ☐ NEW BEST!
 42 ☐ Epoch 031/200 (15.5%) | Train: 85.15% | Val: 84.05% | F1: 0.849 |
 Time: 3s | ETA: 10m 1s ☐ NEW BEST!
 43 ☐ Epoch 032/200 (16.0%) | Train: 85.43% | Val: 84.30% | F1: 0.850 |
 Time: 3s | ETA: 9m 57s ☐ NEW BEST!
 44 ☐ Epoch 034/200 (17.0%) | Train: 85.57% | Val: 84.56% | F1: 0.853 |
 Time: 18s | ETA: 11m 3s ☐ NEW BEST!
 45 ☐ Epoch 036/200 (18.0%) | Train: 86.04% | Val: 85.19% | F1: 0.857 |
 Time: 3s | ETA: 11m 39s ☐ NEW BEST!
 46 ☐ Epoch 038/200 (19.0%) | Train: 86.86% | Val: 85.07% | F1: 0.859 |
 Time: 3s | ETA: 11m 24s ☐ NEW BEST!
 47 ☐ Epoch 040/200 (20.0%) | Train: 86.91% | Val: 85.13% | F1: 0.858 |
 Time: 3s | ETA: 11m 12s
 48 ☐ Epoch 042/200 (21.0%) | Train: 87.35% | Val: 85.85% | F1: 0.866 |
 Time: 3s | ETA: 11m 1s ☐ NEW BEST!
 49 ☐ Epoch 045/200 (22.5%) | Train: 87.55% | Val: 86.10% | F1: 0.868 |
 Time: 3s | ETA: 10m 44s ☐ NEW BEST!
 50 ☐ Epoch 049/200 (24.5%) | Train: 88.45% | Val: 86.51% | F1: 0.871 |
 Time: 3s | ETA: 10m 19s ☐ NEW BEST!
 51 ☐ Epoch 050/200 (25.0%) | Train: 88.73% | Val: 86.08% | F1: 0.868 |
 Time: 3s | ETA: 10m 13s
 52 ☐ Epoch 055/200 (27.5%) | Train: 89.24% | Val: 86.53% | F1: 0.871 |
 Time: 3s | ETA: 9m 44s ☐ NEW BEST!
 53 ☐ Epoch 057/200 (28.5%) | Train: 89.24% | Val: 87.08% | F1: 0.876 |
 Time: 3s | ETA: 9m 33s ☐ NEW BEST!
 54 ☐ Epoch 060/200 (30.0%) | Train: 89.79% | Val: 86.63% | F1: 0.872 |
 Time: 3s | ETA: 9m 17s
 55 ☐ Epoch 062/200 (31.0%) | Train: 89.91% | Val: 87.34% | F1: 0.878 |
 Time: 3s | ETA: 9m 6s ☐ NEW BEST!
 56 ☐ Epoch 065/200 (32.5%) | Train: 90.26% | Val: 87.43% | F1: 0.878 |
 Time: 3s | ETA: 8m 51s ☐ NEW BEST!
 57 ☐ Epoch 070/200 (35.0%) | Train: 90.78% | Val: 86.91% | F1: 0.872 |
 Time: 3s | ETA: 8m 26s
 58 ☐ Epoch 075/200 (37.5%) | Train: 91.32% | Val: 87.37% | F1: 0.879 |
 Time: 3s | ETA: 8m 6s ☐ NEW BEST!
 59 ☐ Epoch 080/200 (40.0%) | Train: 91.70% | Val: 87.34% | F1: 0.879 |
 Time: 3s | ETA: 7m 47s
 60 ☐ Epoch 081/200 (40.5%) | Train: 91.44% | Val: 87.87% | F1: 0.881 |
 Time: 3s | ETA: 7m 43s ☐ NEW BEST!
 61 ☐ Epoch 090/200 (45.0%) | Train: 92.17% | Val: 87.70% | F1: 0.880 |
 Time: 3s | ETA: 7m 9s
 62 ☐ Epoch 091/200 (45.5%) | Train: 92.28% | Val: 87.87% | F1: 0.882 |
 Time: 3s | ETA: 7m 5s ☐ NEW BEST!
 63 ☐ Epoch 095/200 (47.5%) | Train: 92.31% | Val: 87.93% | F1: 0.883 |
 Time: 3s | ETA: 6m 49s ☐ NEW BEST!
 64 ☐ Epoch 096/200 (48.0%) | Train: 92.81% | Val: 88.23% | F1: 0.886 |
 Time: 3s | ETA: 6m 45s ☐ NEW BEST!
 65 ☐ Epoch 100/200 (50.0%) | Train: 92.94% | Val: 88.03% | F1: 0.884 |
 Time: 3s | ETA: 6m 28s


```
106
107     accuracy           0.89    7650
108     macro avg         0.85    0.87    0.86    7650
109     weighted avg       0.90    0.89    0.89    7650
110
111
112     ☐ Model saved as: best_emotion_model.pth
113     ☐ All results saved in 'results/' directory
114
115     Process finished with exit code 0
116
```